



It Does Not Scale Today

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The number: a few sessions per second. [verified] The hosted gateway at `api.mor.org` funds every Web2 caller from one shared consumer wallet. The happy path runs concurrent. After a nonce conflict on a session open or close, on-chain ops serialize on the wallet lock for twenty seconds, and each new conflict re-arms it (`config.py:319-323`). One wallet is one nonce sequence, so under contention the gateway clears a few sessions per second. That is a hard architectural ceiling, not a tuning knob. It is a load test, not an opinion.

A million users, three walls, in order.

- **Wallet.** One nonce sequence, a few sessions per second [verified]. Users do not queue, they collide, and every collision arms the twenty-second lock. There is no admission control and no backpressure; the gateway grep is empty [verified].
- **Supply.** Capacity is `concurrentSlots` per node, dozens, gated at `capacity_manager.go:55`, times 18 providers on `active.mor.org`: a few hundred concurrent sessions [verified]. Most nodes resell Venice and OpenAI keys under their own limits; the node docs show a throttled backend returning 429 [verified].
- **Token.** MOR trades at \$1.58, cap near \$14M [verified]. The stipend is 1% of the pool per day, about 29,700 MOR [derived]; across a million users that is about 0.03 MOR each [illustrative], nothing. So everyone pays direct, and paying direct buys MOR through about \$1.3M of DEX depth, mostly protocol-owned, on about \$57k daily volume [verified].

The fix, named to a repo. Delegated sessions let one funded key authorize many callers, so opens stop serializing on a single nonce. The Delegation facet is already in `Morpheus-Lumerin-Node/smart-contracts (Delegation.sol)` [verified]; LP-320 uses it. Enso, Hanzo's router, puts admission control and backpressure in front of supply [verified]. Settlement moves onto `mord`, where a session is per-session state a validator set orders, not a transaction from one key.

The outcome, measured. [verified: `mord/BENCHMARKS.md`, M1 Max, single core, in-memory, 2026-08-29] `mord` settles the arithmetic in about 2 microseconds per session, roughly 500,000 per second on one core. The un-fakeable cost is two `ed25519` verifies per meter, the provider's and the consumer's, about 296 microseconds, so about 3,400 verified sessions per second on one core. Verification shares no state, so it scales close to linearly: about 25,000 per second at eight cores. The single-wallet wall is gone; the bottleneck becomes how fast consensus orders blocks, which Lux already answers. Honest boundary: 3,400/sec/core is the settlement and verification rate, not end-to-end network TPS, which consensus and networking set. `mord` builds, 40 tests pass; faked work earns zero, a substituted model is slashed [verified].

Relevance. You cannot serve the demand we bring on one wallet and a few hundred slots. The infrastructure has to go horizontal, and ours is. Keep Morpheus thin: MOR stays a budget and rewards token. Inference and proof of work done run on A-Chain (`lux/ai`, `lux/chains/aivm`: bonded operators, commit-reveal quorum, `SettleQuorum`, `SlashProvider`, local attestation, no cloud) [verified]; agentic workloads on the Hanzo agentVM (LP-322 coordination, LP-325 budgets); post-quantum settlement and consensus on Lux primary; governance on Zoo. Morpheus rebuilds none of it; it taps in. An agent runs on the agentVM, verifies inference on A-Chain, and settles budget in MOR.

ADOPT-11. Wall at `config.py:319-323`; throughput at `mord/BENCHMARKS.md`. Fix: LP-320 delegated sessions with Enso admission. Companions ADOPT-01, ADOPT-02, ADOPT-07. Citations checked against the working tree. • research@lux.network